Huawei Access Node Optical Transport Failure Resolution SOP (KOR-ACC-83)

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6724

Description

The failure domain of Optical Transport Failure on the Huawei Access Node (KOR-ACC-83) at Koramangala Datacenter is characterized by a degradation in the performance of the SFP/XFP modules, fiber cuts, or a reduction in Rx/Tx power levels. This is primarily due to the fact that the Huawei Access Node utilizes Optical Transport Network (OTN) protocol for long-haul transmission, which relies on the integrity of the SFP/XFP modules and the optical fibers to maintain signal quality. Furthermore, the Huawei Access Node's design incorporates a distributed architecture, where the control and data planes are integrated within the same chassis, making it challenging to isolate and diagnose issues. The OTN protocol's reliance on the Optical Supervisory Channel (OSC) for monitoring and control adds another layer of complexity to the diagnostic process. As a result, failures in the optical transport domain can lead to a cascade of effects, impacting the overall network performance and requiring meticulous troubleshooting to isolate the root cause.

Pre-requisite Checks

- Ensure access to the Huawei Access Node (KOR-ACC-83) at Koramangala Datacenter is granted and a remote hands team is dispatched to facilitate the troubleshooting process.

- Verify the status of the redundant link and ensure it is operational before proceeding with troubleshooting.

- Review the alarm history and system logs to identify any recent events that may be related to the current failure.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP module performance	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to optical signal degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical interface, Expected Output: Current optical interface parameters, including signal quality and power levels, Description: This command provides an overview of the optical interface's current state, including signal quality and power levels, allowing for the identification of any issues related to the SFP/XFP modules or optical fibers.
```

```
Command: display optical supervisory channel, Expected Output: Current OSC parameters, including signal quality and power levels, Description: This command displays the current state of the Optical Supervisory Channel (OSC), including signal quality and power levels, which can help identify any issues related to the OSC.

Command: display alarm information, Expected Output: Current alarm information, including alarm code and description, Description: This command displays the current alarm information, including the alarm code and description, which can help identify the root cause of the issue.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP module, Command: replace optical interface, Description: This step involves replacing the degraded SFP/XFP module with a new one, ensuring that the new module is properly configured and integrated into the system.

Step 2: Step: Repair or replace the damaged fiber, Command: repair or replace optical fiber, Description: This step involves repairing or replacing the damaged fiber, ensuring that the new fiber is properly configured and integrated into the system.

Post-Check Verification

```
Command: display optical interface, Expected Output: Updated optical interface parameters, including signal quality and power levels, Description: This command verifies that the SFP/XFP module has been replaced or the damaged fiber has been repaired, ensuring that the optical interface is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module or fiber, Command: undo replace optical interface or undo repair or replace optical fiber, Description: This step involves restoring the original SFP/XFP module or fiber, ensuring that the system is returned to its original state in case the fix fails.